AWS – Elastic Cloud Compute – EC2

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Agenda

- What is EC2 instance?
- Types of EC2 instance Families
- Amazon EC2 instance Benefits
- Demo on EC2 instance

Amazon EC2 (Elastic Cloud Compute)

- Resizable compute instances in the cloud
- Provision 1 or many instances
- Pay for what you use; no minimum commitment
- Familiar operating systems, with cloud benefits
- This is a Virtual Machine on AWS.

Components in EC2 Instance

- OS Image
- CPU
- Memory
- Network bandwidth
- Instance Store
- Whether IPv6 is supported or not?

Amazon EC2 Instances - Families

- Compute-Optimized → c4, c5, c5a, c5d, c5n, c6g
- General Purpose → m4, m5, m5a, m5d, m5ad, m6g
- Micro Instances → t2, t3
- Storage-Optimized → i3, i3en,
- GPU Instances → g4dn
- Memory-Optimized → p2, r4, r5 , r5a, r5ad, r5d
- Network Optimized → inf1

Amazon EC2 Instances – Sub Category

- Large → 2 CPU
- xLarge → 4 CPU
- 2xLarge → 8 CPU
- 4xLarge → 16 CPU
- 8xLarge → 32 CPU
- 16xLarge → 64 CPU

Compute-Optimized

Instance Type	Architecture	EBS optimization support	Supported root device types	Network performance
C1	i386, x86_64	unsupported	ebs, instance-store	Moderate

Some Amazon EC2 Benefits

- Elastic/Flexible
  - Use as such as you want and terminate when ever you want
- Completely Controlled
  - Log on as root (Linux) / Administrator (Windows)
  - Create an AMI (Amazon Machine Image)
- Reliable → Use multiple Availability Zones (AZs)
- Secure
  - Secured premises, Storage, Network Access
  - Secured Users , Logins,
- Inexpensive
  - Eg: -- On-demand Instance at 0.02/hr.

Create an simple EC2 instance

- Select an image (OS Image, **ITS NOT AN ISO FILE**)
  - AWS marketplace
  - AWS images
  - Custom images
- Select the instance type . (CPU, RAM and Network Speed)
  - Based on the CPU and RAM size.
  - Custom option is not available, for eg: 3 CPU, 5 GB RAM cannot be configured, instead we would need to select the nearest pre defined types.

Create an simple EC2 instance

- Configure Instance Details
  - Number of Instances
  - Networking
  - Hardware Type (Shared, Dedicated and Dedicated Host)
  - Scale Up feature
  - Terminate Protection → Enable/Disable
  - Configure Network adapter with auto-assign or static private IPv4 and public IPV6.
  - Also has option for additional Network adapter

Create an simple EC2 instance

- Storage Allocation
  - Increase or Decrease the root volume.
  - Type of Disk
    - General Purpose
    - Provisional
    - Magnetic
  - Select the IOPS
  - Enable or Disable “Delete on Termination”
    - This option is **“NOT”** Changeable once the VM(EC2) is created.

Create an simple EC2 instance

- Add Tags → For better manageability.
- Security Groups
  - To provide inbound and outbound Traffic rules for the EC2 instance
- Key Generation
  - It creates an Private and Public key.
  - The private key is downloadable only **ONCE** and AWS does not store it.
  - **Username is default for every IMAGE we select.**

EC2 instance Pricing

Linux | RHEL | SLES | Windows | Windows with SQL Standard | Windows with SQL Web | Windows with SQL Enterprise | Linux with SQL Standard | Linux with SQL Web | Linux with SQL Enterprise

Region: US West (Oregon)

	vCPU	ECU	Memory (GiB)	Instance Storage (GB)	Linux/UNIX Usage
General Purpose - Current Generation					
a1.medium	1	N/A	2 GiB	EBS Only	$0.0255 per Hour
a1.large	2	N/A	4 GiB	EBS Only	$0.051 per Hour
a1.xlarge	4	N/A	8 GiB	EBS Only	$0.102 per Hour
a1.2xlarge	8	N/A	16 GiB	EBS Only	$0.204 per Hour
a1.4xlarge	16	N/A	32 GiB	EBS Only	$0.408 per Hour
t3.nano	2	Variable	0.5 GiB	EBS Only	$0.0052 per Hour
t3.micro	2	Variable	1 GiB	EBS Only	$0.0104 per Hour

Free Services

- 720 hr / Month is free for t2.micro instance type
- 5GB – S3 space is free per month
- 5GB – Glacier Storage
- VPC creation is free

Troubleshooting instance Issues

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Troubleshooting Instance Launch Issues

Troubleshooting Connecting to Your Instance

Troubleshooting Stopping Your Instance

Troubleshooting Terminating (Shutting Down) Your Instance

Troubleshooting Instance Recovery Failures

Troubleshooting Instances with Failed Status Checks

Getting Console Output and Rebooting Instances

Booting from the Wrong Volume

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-instance-troubleshoot.html

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